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Analysis of the viability of a giant panda population

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Summary

1. A model was set up to simulate the dynamics of a giant panda *Ailuropoda melanoleuca* population. All parameters in the model were taken from an existing panda population in the Shuidonggou Valley area of the Qinling Mountains, China.
2. The model projects population dynamics and assesses population viability under changing conditions.
3. Small, fragmented populations face potential risks in terms of genetics and demographic processes. This study concerns the latter.
4. The simulation indicated that: (i) the population is stable under present conditions (but note that the model does not incorporate environmental stochasticity); (ii) the environmental carrying capacity is the most important factor restricting population increase; (iii) poaching would greatly affect the probability of population survival; (iv) increasing the survival of individuals under 1 year of age would increase population persistence substantially; (v) reducing subadult female mortality would accelerate population growth; (vi) shortening the interbirth interval would allow more rapid population growth, and its effect is greater than changing other parameters, but may not be practicable; and (vii) immigration and reduction in young male mortality are of limited benefit to this small population but may be important in practical conservation management.
5. The model suggests management strategies for giant panda conservation.

Key-words: conservation, Leslie matrix models, poaching, population modelling, population viability.

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Introduction

In recent decades, the numbers of wild giant panda *Ailuropoda melanoleuca* have decreased rapidly. This is mainly due to the destruction and fragmentation of their natural habitat by logging, which has separated a once well-integrated giant panda population into many sub-populations of small size (Schaller *et al.* 1985).

Small populations often suffer a loss of genetic diversity caused by inbreeding, and are susceptible to further reduction in numbers through continuing habitat loss and poaching. It is of critical importance to understand the dynamics of these small populations to determine whether such populations are likely either to go extinct or to persist for extended periods.

Research scientists have used various models to project the dynamics of different species. However, minimum viable population (MVP) size often remains unclear because the value for MVP depends on the type of model used. Simulation of real populations, especially those which are already isolated, may be a good alternative method.

In recent years, much research has been done on small populations, including wild Mauritius kestrel (Temple 1986), acorn woodpecker (Stacey & Taper 1992) and saiga antelope (Milner-Gulland 1994). Our study uses Leslie matrix models to project the population dynamics (Smith & Trout 1994) of a small group of giant pandas. The models are based on population parameters collected in the field over 10 years. Data on the internal genetic diversity of this reproductive group are lacking so we have not considered the influence of inbreeding here.

Methods and field observations

STUDY AREA

The Qinling Mountains lie in the middle of China, adjacent to the Tibetan Plateau on the west and the

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Ba Mountains to the south. This massive mountain range divides the subtropical zone from the semi-arid zone. Because of the notable differences in geography and climate between the southern and northern slopes of the range, the Qinling Mountains form the natural boundary between the Oriental realm and the Palaearctic realm in the middle of China.

Our study was concentrated in an area of about 140 km^2, located in the middle part of the south slope of the Qinling Mountains, near Huayang Town (33°33′ to 33°45′ north latitude and 107°30′ to 108°00′ east longitude) and upstream of the Youshui River (one branch of Hanjiang River). Xinglongling Ridge, the second important ridge of the Qinling Mountains, formed the northern boundary, and the 1100 m countour, the lower limit of pandas' activity, was regarded as the southern boundary. The study group of pandas occupied an area of about 12 km^2. This area included Shuidonggou and five other valleys and was bounded by the Hetaoba River to the south and the Cang'eryan River to the west (Fig. 1).

The study area included two climatic zones: the mountainous warm temperate zone and the alpine cool temperate zone. Correspondingly, there were three principal types of vegetation: temperate deciduous broadleaf forest, mid-mountain theropence drymion forest and subalpine needleleaf forest. The climate was temperate and moist. There were about 140–160 frost-free days per annum. Annual precipitation was greater than 1000 mm and the average annual temperature was 9 °C.

Two kinds of bamboo were present, *Bashania fargesii* and *Fargesia spathacea*. *B. fargesii* grew from 800 to 2100 m and provided pandas with winter and spring food. *F. spathacea* grew from 1900 to 3100 m, and supplied the majority of summer and autumn food for pandas. No large-scale bamboo flowering has occurred in this district in the past 200 years so bamboo flowering was not considered in the model. Shuidonggou Valley belongs to the people living there and the volume of standing timber was much lower than surrounding areas because of local timber cutting.

During the study period, a few pandas were poached in nearby areas but this never happened to the group under research. The usual cause of death was disease or old age.

CAPTURE AND RADIO-TELEMETRY

Field research has been carried out for the last 10 years (1985–94), with at least 8 months of each year spent in field study. Pandas were tracked in snow in winter and then anaesthetized with ketamine hydrochloride solution (170 mg ml^{-1}; $\approx$ 4 mL per 100 kg body weight) injected from a dart gun. The panda's teeth, fur, limbs, claws and genital organs were carefully inspected, its age was estimated, and it was measured and searched for parasites. Pandas were fitted with radio-collars (Telonics Company, MOD-500 Telemetry Collar, 150–152 MHz, 0·5–1 kg weight, battery life 2 years and 8 months), allowed to fully recover from the anaesthetic and released. A few pandas removed their collars but the remainder were radio-tracked for between 1 and 60 months (Table 1).

Individual pandas were identified by the unique frequency of the radio-collar and/or by individual body characteristics (e.g. ear nick, wound on nose, speckles on face).

In March and April every year, breeding behaviour was observed at the mating site. In August, a pregnant female panda would usually give birth in a secret den. Using radio-telemetry it was often possible to find the mother and baby. More than one den was used every year and the same panda would often reuse a den.

Since 1985, 21 individuals have been captured and radio-collared in the study area (see Table 1).

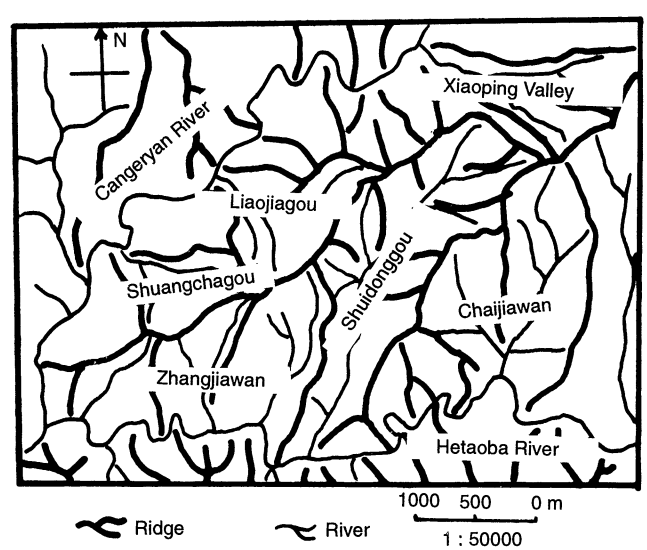

Fig. 1. The study area with locations.

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Table. 1. Details of radio-collared pandas in the Shuidonggou group

Panda	Sex*	Age (years)†	Period of being collared (month)	Total	Notes
Jiaojiao	F	8·5	11 Mar 1989 – now‡	60	
Xiwang	F	1·5	Aug 1993 – now	8	Observed since birth
Huzi	M	4·5	29 Jan 1991 – now	38	Observed since birth
Ruixve	F	10	26 Mar 1993 – now	12	
Baoma	F	8–10	Mar 1990 – Mar 1992	24	Collar lost
Dabai	M	10–13	Mar 1992 – now	24	
Keke	F		Feb 1992 – May 1993	16	Battery dead
Meimei	F	6–7	31 Mar 1993 – now	12	
Jiaoshou	M	5·5	17 Feb 1993 – now	13	
093	M		13 Mar 1992 – now	24	
094	M		31 Mar 1993 – now	12	
Boshi	F	3·5	19 Feb 1993 – now	13	
Daxia	F		24 Feb 1993 – now	13	
Yanghe	F	Dead	10 Feb 1989 – Feb 1990	13	Dead in 1991 at 17 years old
Daxiong	M	Dead	Apr 1990 – Apr 1991	13	Dead in 1991 at 17 years old
Xiao F	F	?	Jan 1990 – Mar 1990	3	Collar lost
Shuilan	F	Dead	2 Feb 1991 – Dec 1991	11	Dead in 1991 at 4 years old
Shanshan	M	Dead	16 May 1987 – Mar 1988	11	Dead in 1988 at 13–14 years old
1331	M		1993 – ?		
Xiaxia	F		28 Sep – 24 Oct 1989	1	Old
005	M		Feb 1990 – ?		

*F = female, M = male.
†Ages were estimated as of March 1994.
‡As at March 1994.

DEFINITION OF THE PARAMETERS USED IN THE MODEL

The reproductive group of pandas studied occupied an area of about 12 km^2 in winter and spring. This area (including Shuidonggou and five other valleys) was bounded by the Hetaoba River to the south, and the Cang'eryan River to the west (Fig. 1). Members of this group migrated seasonally, generally spending June to September in *Fargesia* forest (2200–3000 m), referred to as summer habitat, and from September to May foraging in *Bashania* forest (1300–2100 m) referred to as winter habitat.

Giant pandas generally live alone and were thought to constitute different reproductive groups in winter based on the following evidence (Lu Zhi 1992).

1. Adults of both sexes had relatively stable home ranges. The home range of one male usually encompassed the home ranges of several females. This male usually had the most mating opportunities.
2. Although the home ranges of females overlapped, they all had stable exclusive core areas.
3. The home ranges of females were determined primarily by the distribution of food resources, while those of males were determined by the number and availability of reproductive females.
4. The strongest male held the dominant position in competition for mates. Young males were nomadic between different groups whilst old individuals ceased breeding.
5. Subadult females dispersed but subadult males were resident.
6. A reproductive group used the same site for mating each year.

Apart from one transient female (Xiao F), two resident-born females who later dispersed (Boshi and Shuilan), and one individual belonging to an adjacent group (Xingxing), all other radio-collared pandas were resident individuals belonging to the Shuidonggou group. In addition, data was used from three radio-collared pandas (043, Qingqing and an unnamed female) from a neighbouring district (Foping County). In addition, data from zoos was occasionally used to supplement the field data (Zhao Qingguo *et al.* 1993).

The model parameters are discussed below.

Reproductive interval

In the model the reproductive interval was defined as 2·5 years (see Table 2).

Age of maturity

Data from this study indicate that female pandas in Qinling mature at about 3·5 years and males at 4·5 years.

Maximum breeding age

During this study, the male Daxiong mated in 1989 and died in 1991(Table 1) at about 17 years of age. In Xi'an Zoo, a male named Wanwan mated when he

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Table 2. Reproductive data from two pandas

Panda	Time of mating	Time of giving birth	Time of infant leaving mother	Reproductive interval
Jiaojiao	March 1989	August 1989	March 1992	3 years
Jiaojiao	March 1992	August 1992	March 1994	2 years
Jiaojiao	10 April 1994	August 1994		
Ruixve	March 1990	August 1990	March–April 1993	2 years
Ruixve	March 1993	August 1993		

was 16 years old. Based on this evidence, the maximum age of breeding for males was considered to be 16 years (Pan Wenshi, Lu Zhi & Zhu Xianjian 1993).

This study yielded no conclusive data on the maximum breeding age of females in the wild. In 1990, Yanghe mated but did not give birth the following autumn. She died in 1991 aged about 17 years (Table 1). In Beijing Zoo, a female panda named Daidai was pregnant at 19 years of age. For the purpose of the model, the maximum breeding age for females was considered to be 19 years.

Survival

Estimates of survival (Table 3) were obtained from the 24 collared pandas and their cubs. This included 10 females who were 4–18 years old: Dandan, Fangfang, Yanghe, Jiaojiao, Ruixve, Baoma, Meimei, Keke, Daxia and an unnamed collared panda in Foping Reserve. Yanghe died in 1991 aged 17 years. Fangfang died of cancer aged between 14 and 15 years. Dandan contracted enteritis aged 10 years and became unable to survive in the wild without help. She was rescued and now lives in Xi'an Zoo. The other individuals were still alive in March 1994, thus the annual survival was about $\sqrt[15]{7/10} = 0{\cdot}977$ for ages 4–18 years.

Field observation of the fate of known individuals yielded further information on some of the factors contributing to survival.

1. Direct evidence of subadult female dispersal came from Shuilan and Boshi. They both disappeared from the study group. Shuilan moved 15 km away but was unable to establish herself in a new area and she returned after 4 months of nomadism. She died several months later. By contrast, Boshi successfully moved to a new area about 10 km away.

No direct evidence of immigration was acquired in this study. Xiao F was thought to be a transient subadult. She was radio-collared in the home range of Jiaojiao (a resident female adult) but was forced to flee the area after an intense struggle lasting over an hour. Xiao F slipped her collar a few days later.

The records of five subadult females (Shuilan, Boshi, Qingqing, Xiao F and 043) indicated that mortality was high (see Table 1).

2. The majority of males had wounds from struggles and fights. Some adults (e.g. Xingxing) who consistently lost fights stopped breeding altogether although still apparently of reproductive age. For the purposes of the analysis they were considered 'dead'. Males aged 4–6 years were vulnerable to wounding in territorial disputes. At this age they were just beginning to establish their own home ranges but they were inexperienced and not strong enough to defeat older resident males. Their mortality was relatively high.

3. No deaths were recorded of males between 1 and 3 years of age or of female yearlings. At that age they still lived with their mother and received good protection and nutrition. Their survival was considered to be 100%.

Table. 3. Survival of each age group of pandas

	Survival from age j to $j + 1$ years	
Age group	Females	Males
0	0·833	0·833
1	1	1
2	0·707	1
3	0·707	1
4–6	0·977	0·928
7–15	0·977	0·980
16	0·977	0
17–18	0·977	/
19	0	/

Population size and composition

Based largely on the identity of pandas at breeding aggregations in 1993 and 1994, the number of pandas in this reproductive group was estimated. There were nine adult famales, including the six listed in Table 1 (Jiaojiao, Baoma, Meimei, Ruixve, Daxia, Keke), one unnamed individual in Shuidonggou (repeatedly observed), and two unknown individuals in Chaijiawan and Zhangjiawan, respectively. The main areas of activity of the six radio-collared female pandas are shown in Fig. 2. The winter and spring forage was sufficient in two other areas to support females, hence the existence of two unknown females was deduced.

There were seven adult males in the study area including those in Table 1 (Dabai, Huzi, Jiaoshou, 093, 094) and two unnamed individuals observed in

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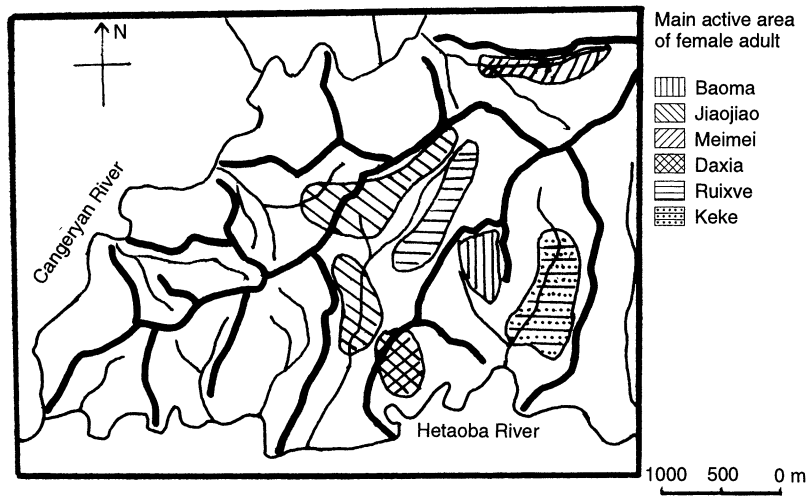

Fig. 2. The main activity areas of adult females in 1992 and 1993.

March 1993. In addition, two unnamed old individuals were observed.

The remaining nine study animals included cubs (still dependent on their mother) and subadults. These were: Xiwang, two infants of Ruixve (0·5-year-old and 3·5 years old), an infant of Daxia, an infant of Boma (1·5 years old), an infant of Keke, and three infants of the three unnamed individuals.

In summary, it was estimated that this reproductive group comprised nine mature females, seven mature males and nine cubs and subadults. Including the two old pandas, the population size in late 1994 was 27. The model simulations concerned this population.

Environmental carrying capacity of this area

The reproductive group studied appeared to have the highest density of any group in the Qinling Mountains. All suitable space appeared to be occupied by the nine females. Most of the time, each female occupied a relatively exclusive and stable active area with no vacant areas inbetween. This is borne out by the observation that, in August 1993, Ruixve gave birth just 50 m from the active area of Jiaojiao and Xiwang (daughter of Jiaojiao). Similarly, the males appeared to have occupied all available habitats. Thus, the observed density of pandas was defined as the environmental carrying capacity.

SETTING UP THE MODEL

The model is concerned only with individuals that have potential reproductive capacity, including cubs, subadults and adults. Old pandas that were no longer able to breed or bear cubs, and hence had no influence on future population size, were ignored. All parameters in the model relate to that sector of the population which is potentially reproductive, e.g. population size, numbers of females, numbers of males.

According to a Leslie matrix (Smith & Trout 1994), if N_i is the total population size of the ith year, n_{ij} is the number in the jth age group in the ith year, and x is the maximum age of reproduction, then

$$N_i = \sum_{j=0}^{x} n_{ij}$$

Therefore, $n_{ij} = n_{(i-1)(j-1)} \cdot S_{(j-1)}$, while $S_{(j-1)}$ is the survival of individuals from $(j-1)$th to jth age group.

Since the minimum and maximum age at reproduction was different for the two sexes, the size of every age group was calculated separately.

Initially, no immigration or emigration was included.

The sex ratio at birth was assumed to be 1:1 (Schaller *et al.* 1985).

Since it was impossible for the number of reproductive individuals to exceed the environmental carrying capacity, any surplus was removed at the end of each year.

If the number of either sex declined to less than 2, we considered the population extinct (Huang Chengming & Hu Jinchu 1989; Xia Wuping & Hu Jinchu 1989).

Results of the modelling exercise

Under current conditions the population will remain fairly stable

If poaching or other factors are not considered, and if immigration is considered to be equal to emigration, the population will persist for over 200 years (Fig. 3). The demographic parameters of the population are presented in Table 4.

Effect of environmental carrying capacity

The size of the population at equilibrium was related to the environmental carrying capacity (Fig. 4). When the carrying capacity was doubled in the model, the

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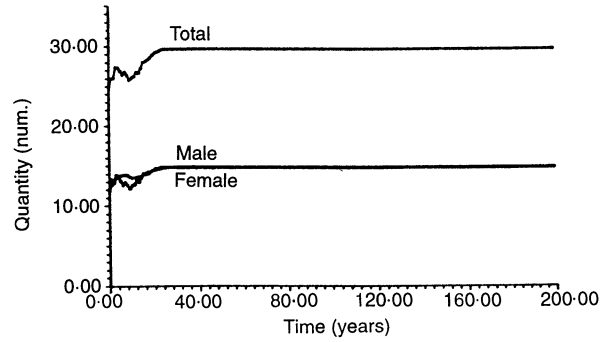

Fig. 3. Dynamics of the study population over 200 years.

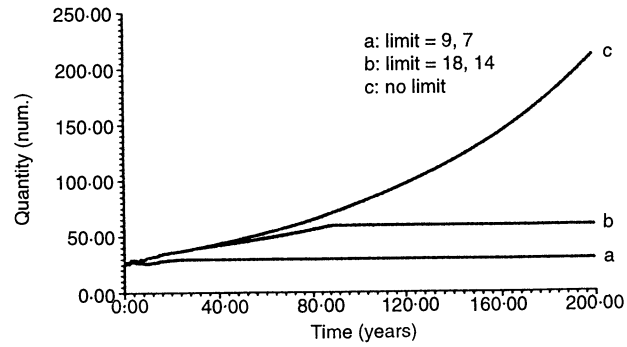

Fig. 4. Effect of environmental carrying capacity.

Table 4. Predicted population parameters after 200 years

Parameters	Mean	Variance	Stable value
Population size	29·39	0·87	29·66
Number of females	14·70	0·50	14·86
Number of males	14·68	0·39	14·80
Population sex ratio	1·00	0·017	1·00
Adult sex ratio	1·27	0·099	1·29
Proportion of young in the population	0·46	0·011	0·46
Proportion of adults in the population	0·54	0·011	0·54
Fecundity	0·12	0·0089	0·12
Mortality	0·12	0·01	0·12
Rate of population increase	0·00087	0·0062	0·00

population increased for 87 years, with an average annual rate of increase of 0·98%, and then remained stable. When the carrying capacity had no upper limit, the population continued to increase at a rate of 1·08% per annum, and reached 236·88 after 199 years.

Effect of poaching

The probability of the population going extinct at different levels of poaching are displayed in Table 5 and Fig. 5. It can be seen that the population would suffer dramatically from poaching, even at rates as low as 0·3 pandas poached per year. A poaching rate of 0·5 pandas per year would cause the population to go extinct. Such a small population is very vulnerable and needs to be protected carefully.

Effect of immigration

The model was used to simulate the effects of immigration into a small population by a few individuals

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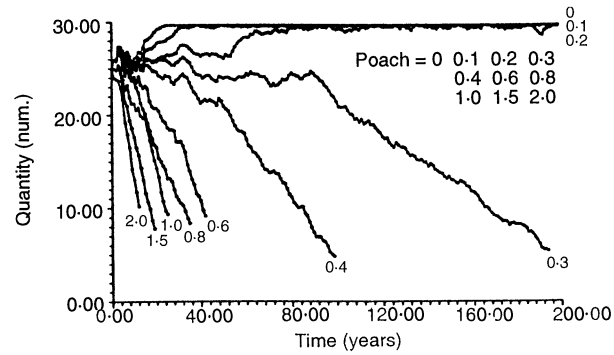

Fig. 5. Effect of poaching (or removal).

Table 5. The relationship between poaching and the probability of extinction

Number poached year^{-1}	Average time to extinction (years) *	Probability of extinction In 100 years	Probability of extinction In 200 years
0	/	0·00	0·00
0·1	/	0·00	0·00
0·2	/	0·00	0·00
0·3	/	0·6	35·8
0·4	/	52·9	99·1
0·5	68·65 ± 17	95·5	100·0
0·6	52·20 ± 12	99·8	100·0
0·7	41·78 ± 8·7	100·0	100·0
0·8	35·69 ± 8·1	100·0	100·0
0·9	31·08 ± 7·4	100·0	100·0
1·0	26·06 ± 5·8	100·0	100·0
1·5	16·79 ± 3·5	100·0	100·0
2·0	12·24 ± 2·1	100·0	100·0

*'/' indicates that the population will persist after 200 years.

each year, e.g. where pandas born in zoos might be returned to the wild (Fig. 6). Immigrant pandas could belong to any age group. An immigration rate of one, two or three pandas per year into an unsaturated population elevated the rate of population increase to 1·12%, 1·16% and 1·17%, respectively, compared to a rate of increase of 1·08% for a population with no immigration.

Effect of reproductive cycle

The simulation showed that if the reproductive cycle (inter-birth interval) was reduced to 2 years, the population would increase very quickly at a rate of 3·03% per annum (Fig. 7). If the reproductive cycle was extended to 3 years, the population declined. If it was extended beyond 3 years, the population rapidly went extinct.

Changes in mortality

(i) *Increase or decrease in the survival of the first-year age group.* In the field, the survival of cubs from birth to 1 year old was 83·3%. When cub survival in the model was increased to 90%, the population grew more quickly, with an average annual rate of increase in 200 years of 1·47%. When cub survival was reduced to 70%, the population declined at a rate of −0·37% per annum (Fig. 8).

(ii) *Increase in the survival of subadult females.* When the survival of subadult (2–3-year-old) females was increased to 75% and 80%, the annual rate of population increase reached 2·07% and 3·19%, respectively. Clearly, increasing the number of females with reproductive potential would eventually lead to increased fecundity (Fig. 9).

(iii) *Increase in the survival of subadult males.* When the survival of subadult males belonging to the 4–6-year-old age groups was increased from 92·8% to

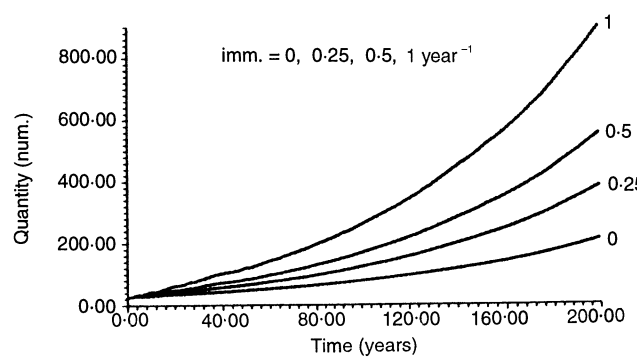

Fig 6. Effect of different immigrant numbers.

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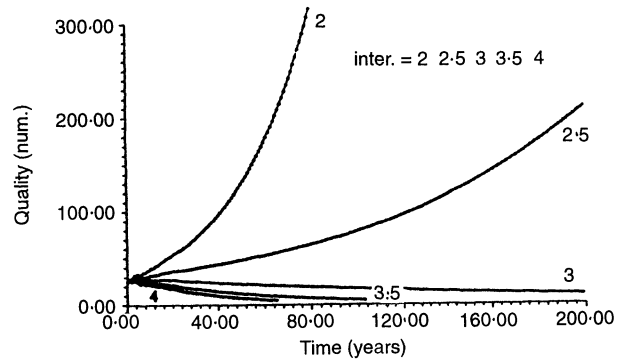

Fig. 7. Effect of female reproductive interval.

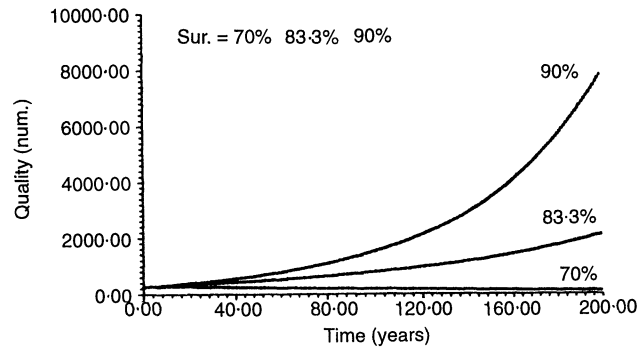

Fig. 8. Effect of survivorship in the first-year age group.

95%, the population dynamics in the model did not show any obvious change. The rate of population increase rose slightly from 1·08% to 1·09% (Fig. 10). Comparison of Figs 9 and 10 shows that the effect of increasing the survival of subadult females was much greater than increasing the survival of subadult males.

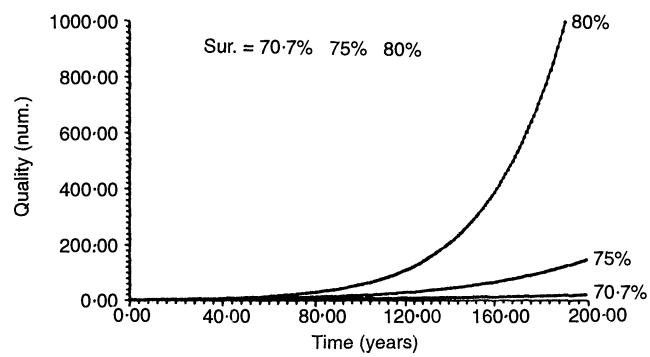

Fig. 9. Effect of survivorship of subadult females.

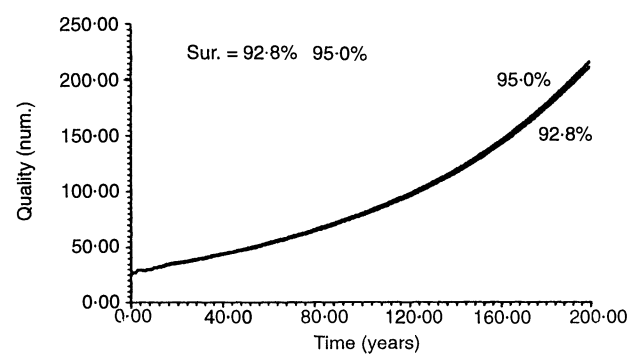

Fig. 10. Effect of survivorship of subadult (4–6-year-old) males.

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In the model, male pandas are considered to have little effect on population growth.

Extending the reproductive age range of adult females

Extending the reproductive age range of females had a strong effect on population growth (Fig. 11). An extension of 1 or 2 years resulted in a population increase of 1·47% and 1·82% per year, respectively.

Effect of increased mortality

When annual mortality was increased by just 3% the population became extinct in 200 years; higher rates of annual mortality resulted in more rapid extinction (Fig. 12). This 'extra' mortality resulted in a rapid reduction in the average time to extinction (Fig. 13) and a rapid increase in the likelihood of extinction (Table 6).

Discussion

THE FUTURE OF THE POPULATION

The results of the simulation modelling indicate that under current environmental conditions the giant panda population is relatively stable and will remain so as long as those conditions do not change. The model does not, however, incorporate the effects of environmental stochasticity on demographic parameters, which may strongly influence extinction probabilities (Dennis, Mulholland & Scott 1991; Milner-Gulland 1994). Therefore, the apparent stability must be monitored closely and cannot be taken for granted.

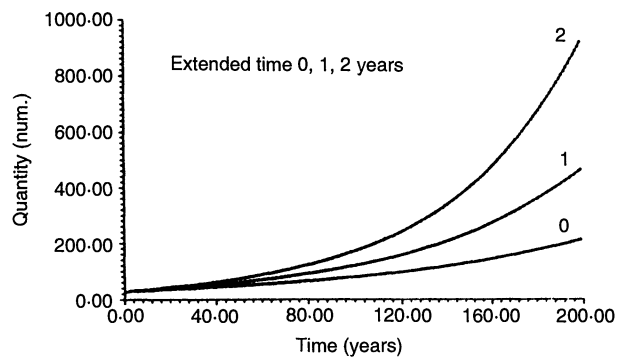

Fig. 11. Effect of extending the reproductive age range of females.

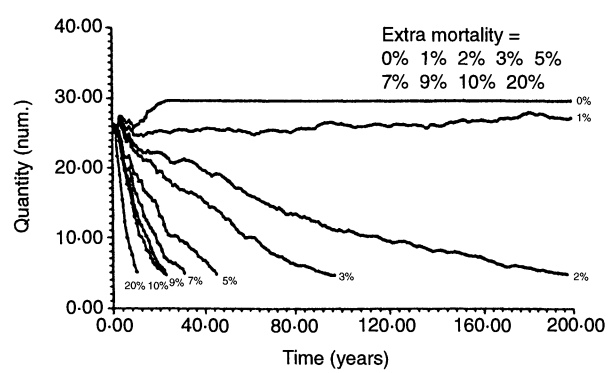

Fig. 12. Effect of extra mortality.

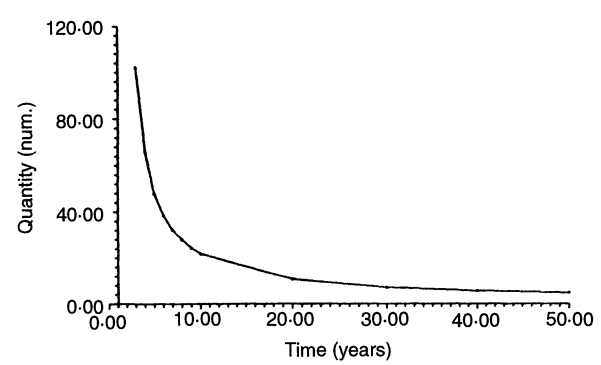

Fig. 13. Relationship between extra mortality and time to extinction.

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Table 6. Likelihood of extinction with increasing annual rate of mortality. The results are based on 1000 stochastic simulations

Increase in annual mortality (% year^{-1})	1	2	3	> 4
Probability of extinction in 100 years	0·00	0·00	33·6	100·0
Probability of extinction in 200 years	0·00	2·1	100·0	100·0

PROBLEMS FACING THE GIANT PANDA

One major threat to giant panda survival is loss of, and damage to, habitat. Most parts of the south slope of the Qinling Mountains were controlled by several timber companies – a typical feature of giant panda habitat. The core area of the pandas' range was occupied by four timber companies. This area was logged repeatedly and replanting usually resulted in all bamboos and bushes being cut down and replaced by monocultures. For example, in recent years large areas have been planted with Japanese deciduous pine by the Huayang Timber Farm. The consequent food shortage forced pandas to emigrate to adjacent habitats with a concomitant rise in panda density therein. In turn, this led to more frequent fighting between male pandas, increased competition among subadults seeking a vacant area to settle, and increased competition between breeding females for caves or other dens suitable for raising cubs.

A second major threat to giant panda survival is from poaching. The Chinese government has forbidden the capture of giant panda from the wild for many years; this policy must be strictly enforced continuously.

It is crucial to the survival of the species in the wild that the remaining wild populations are protected from poaching and further habitat loss or degradation.

CONSERVATION MANAGEMENT

Increase the environmental carrying capacity

Environmental carrying capacity was found to be an important factor restricting population size. As human activity reduces the habitat available to the population so the environmental carrying capacity is reduced and population size declines (Lovejoy & Oren 1981). In fact, this was already happening in the area adjacent to Shuidonggou.

One method of increasing environmental carrying capacity is to restore habitat. Damaged landscapes should be restored as soon as possible. Logging should cease, and humans should emigrate from giant panda habitat. Historically, the study area had been agricultural land and many valleys still had some trace of agriculture. However, every time humans moved away, bamboos and trees regrew and pandas immigrated back. Thus, habitat restoration can effectively accelerate population growth (Rogers 1987).

Shorten the reproductive interval of females

The model suggests this was the most effective means of increasing the size of the population. However, we do not know why female pandas have cubs every 2–3 years. Female experience in nursing cubs, the health of mothers, or changes in hormone levels may all affect inter-birth intervals. We do not recommend artificially removing cubs from their mothers, as this may increase cub mortality. Living with their mother may be an important learning process for cubs, with direct influence on their future life. Thus, although this was the most effective method in the model, we urge caution before extrapolating to actual management action.

Extend the reproductive age

Wild giant pandas are polygynous. Male pandas gain their status and mating rights in intra-sexual competition. Female pandas have exclusive core activity areas, and sometimes showed territorial behaviour (Lu Zhi 1992). Therefore, it would be difficult to extend the length of reproductive age with wild pandas. This might be possible for pandas, especially females, in zoos where artificial insemination and other methods may assist successful pregnancy.

Minimize subadult female mortality

The model indicated that it was quite effective to increase the survival of subadult (2- and 3-year-old) females. Adding only females was more effective than adding both sexes to a polygynous population. In the wild, dispersing subadults face two types of difficulties. The first is being exiled by resident females. For example, female adult Jiaojiao and subadult Xiao F fought together. There may be less conflict if habitat availability increased and panda density was reduced. The second difficulty is physical barriers to dispersal. Agriculture, logging, tree plantations and road building have all destroyed corridors between higher elevation habitats, and the panda range has become fragmented. These barriers are likely to increase mortality during dispersal. Therefore, we must protect existing habitat corridors, and rebuild lost corridors, to facilitate subadult dispersal (Burkey 1989).

Minimize young male adult mortality

Although, according to the model, this was not an effective method of increasing overall population size,

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if more males can mate there is a better chance of preserving genetic variability in a small population.

Raise survival of the first-year age group

We believe that carnivores in the Qinling Mountains, including black bears *Ursus thibetanus* and Indian martens *Martes flavigula*, are the major danger for 0–1-year-old cubs (Pan Wenshi & Lu Zhi 1988). Field research indicated that experienced mother pandas were more successful at raising a cub. For example, Jiaojiao successfully raised three cubs, whilst the first cub of Meimei died in 6 months. Well hidden den sites can also improve cub survival. Thus, the most favourable caves for mother pandas in this region should be identified through survey and then protected. Such protection would include stopping logging, hunting and other human activity in the forest near these sites.

Encourage immigration

The model reflects the fact that adding a few individuals will have only a small effect on overall population growth (in practice, the number of immigrants could only be small). Nevertheless, immigration can be extremely important in two respects. First, this population and others nearby might constitute a metapopulation, in which case immigration could counteract the effects of poaching and environmental stochasticity. Secondly, immigration would increase genetic flow between different populations, reducing the (unknown) risks of inbreeding. In the study, one reason for the increase of the Shuidonggou group was the immigration of animals as a result of damage to surrounding habitats. The reintroduction of pandas from zoos is also a potential method of immigration.

DEFICIENCIES OF THE MODEL

1. The parameters used in Leslie matrix projections are constant. However, in reality some of them will change over time. For example, the mortality of young males might decrease if the density of adult males decreased. Similarly, the survival of immigrating subadult females may change with resident female density. Reproductive intervals are also not constant. We lacked data to evaluate the variance in parameter values.

When simulations proceed for short periods, the errors will be small. Over longer periods, the difference between prediction and reality may be larger. However, the results of such models are still valuable because they illustrate trends in populations.

2. The sample sizes for all parameter estimations were small. For logistical reasons, only 21 pandas were collared. This problem was partly remedied by using data from captive pandas.

3. Environmental stochasticity was not reflected in the model. We lacked data on the effects of environmental variation on demography.

4. The influence of genetic variability was not considered. We had no information on the genetic diversity of a wild giant panda population. We did, however, set up a parameter labelled 'extra mortality' and this might be used to evaluate the influence of inbreeding depression, if this is ever quantified.

Conclusion

Researchers have used Leslie matrix models in recent years to study herbivores, e.g. white-tailed deer (James & Wiegert 1994), saiga gazelle (Milner-Gulland 1994), rabbits (Smith & Trout 1994) and birds, e.g. acorn woodpeckers (Stacey & Koening 1984). All these species have very different life histories from giant pandas. Despite this, all these studies concluded that small populations are fragile, and that small changes in key parameters can lead to extinction. For the panda, it is clear that scientific management and conservation are crucial for its survival.

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